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PPASurrogate: Post-Route Surrogate-Guided Placement Configuration Optimization

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Summary

This paper introduces PPASurrogate, a placer-agnostic framework that learns a post-route PPA surrogate from mixed-size placement states and uses it to guide configuration optimization via Bayesian search. A careful label fidelity study across 10 ChiPBench designs shows post-global-routing (GRT) labels are far more predictive of final PPA than HPWL or pre-route STA, motivating a dual-stream predictor (CNN over spatial density/congestion grids + GAT over macro connectivity) trained on post-GRT labels. On five unseen designs, surrogate-guided BO over DREAMPlace hyperparameters improves post-route WNS and TNS by 12% and 21% vs. Hier-RTLMP within 50 trials, while prior AI placers remain above the baseline; the surrogate also transfers across placers (DREAMPlace→RTLMP) without retraining.

Strengths

Technical novelty and innovation

- The paper isolates and validates post-GRT as a cost-effective, high-fidelity supervision stage for cross-stage learning, replacing widely used but low-fidelity pre-route STA.
- A dual-stream mixed-size representation (spatial CNN + macro-level GAT) is a principled step beyond macro-only encodings, and ablations show complementary gains from both streams.
- Placer-agnostic design: the surrogate acts on finished placements, enabling transfer across fundamentally different placement paradigms without retraining.
- Integration with BO over placer hyperparameters is simple yet effective, demonstrating how a learned PPA surrogate can practically drive better placements.

Experimental rigor and validation

- Label fidelity study over 10 diverse circuits systematically quantifies Spearman correlations at multiple stages (HPWL, pre-CTS, post-CTS, post-GRT, post-DRT) with runtime costs.
- End-to-end PPA evaluation on five held-out circuits with multiple runs, plus extensive ablations (label stage, representation, architecture) and BO convergence analysis.
- Cross-placer transfer test (DREAMPlace-trained surrogate scoring RTLMP placements) shows meaningful generalization.

Clarity of presentation

- The problem is well-motivated by recent benchmarks highlighting the HPWL–PPA disconnect; the pipeline is clearly depicted (offline training, online BO/selection).
- Mathematical formulation and data representations are explained with sufficient detail to reproduce the main components.

Significance of contributions

- Demonstrates a practical path to surpass a strong hierarchical macro placement baseline on post-route timing, addressing a critical gap reported by ChiPBench.
- The label-fidelity analysis provides actionable guidance to the community on supervision choices for cross-stage predictors.

Weaknesses

Technical limitations or concerns

affect conclusions.

- The ranking loss uses a dot-product over z-scored metrics after sign unification, implicitly assuming co-monotonic improvements across objectives; this may not reflect real trade-offs (e.g., timing vs. power) and could bias training.
- Area is constant per circuit but is included as an output/term; this risks adding noise to training or complicating rank loss without benefit.

Experimental gaps or methodological issues

- Missing head-to-head comparison to AutoDMP (BO over DREAMPlace with proxy objectives) under the same evaluation budget, which would directly quantify the impact of replacing HPWL/congestion proxies with the learned PPA surrogate.
- Fairness of LaMPlace comparison relies on published numbers and partial baseline verification (two designs), leaving possible tool/version drift unaddressed for the other test circuits.
- Offline data generation cost is undercharacterized; training reportedly uses ~5,000 post-GRT labels but runtime/resource requirements and scalability to larger tech nodes/designs are not fully quantified.

Clarity or presentation issues

- The objective scalarization for BO and the handling of multi-metric tradeoffs (e.g., lexicographic vs. weighted sum vs. constrained optimization) are not specified.
- Some notation inconsistencies and typos (e.g., “PWNS/PTNS” in tables) and occasional ambiguity around “mixed-size” scope (macro vs. standard-cell specifics in the GAT) could be cleaned up.

Missing related work or comparisons

- Multi-fidelity BO/active learning strategies (occasionally validating with real post-GRT/DRT to recalibrate the surrogate) are not explored as baselines or ablations.

- Recent transfer or surrogate-based placement tuning frameworks beyond LaMPlace and AutoDMP (e.g., broader "shift-left" VP/FA hybrids) could be more thoroughly positioned.

Detailed Comments

Technical soundness evaluation

- The identification of post-GRT as the best supervision stage is strongly supported by consistent, high Spearman ρ across 10 circuits and plausible runtime savings vs. post-DRT. This is a valuable and well-executed empirical result.
- The dual-stream architecture is well-justified: spatial grids encode density/congestion patterns (RUDY, pin density, net-BB density), while GAT captures macro connectivity and geometry; ablations show both streams contribute and combine beneficially.
- The ranking loss is appropriate for the intended use (configuration ranking), but the chosen dot-product multi-metric form could suppress useful trade-off information. A scalarization sensitivity study (e.g., weighted sums, Pareto classification) would strengthen the case.
- The BO formulation and fixed-seed strategy to deconfound seed noise from configuration quality are sound; the convergence and fANOVA analyses increase confidence in the optimization dynamics.

Experimental evaluation assessment

- The label fidelity study is a clear highlight: it is broad (10 circuits), reports both timing metrics, includes runtime implications, and contrasts multiple stages and proxies.
- End-to-end results on five held-out circuits show consistent timing improvements below the hierarchical macro baseline with modest run budgets (50 trials), which is practically significant.
- Ranking accuracy metrics (Spearman/Kendall, Top-k hit rates) are appropriate, and cross-placer transfer results (ρ≈0.63 for WNS) are compelling evidence of placer-agnostic behavior.
- However, a direct comparison to AutoDMP under equal budgets would isolate the benefit of replacing HPWL/RUDY proxies with a learned PPA surrogate in BO. Similarly, a multi-fidelity BO variant (e.g.,

infrequent post-GRT validations during BO) would contextualize the single-surrogate approach.

- Runtime accounting is helpful for inference/BO, but the offline data generation and model training costs should be reported more completely (cores/GPUs, wall-clock, parallelism assumptions) to assess adoption feasibility.

Comparison with related work (using the summaries provided)

- ChiPBench (2407.15026) documented weak HPWL/MacroHPWL correlation with timing and emphasized end-to-end PPA evaluation; this work operationalizes that insight by building a robust post-GRT surrogate and validating end-to-end gains over a strong baseline.
- AutoDMP (ISPD'23) uses BO with HPWL/congestion objectives; PPASurrogate's main advance is replacing weak proxies with a data-driven, high-fidelity surrogate aligned to timing. A direct experimental juxtaposition would sharpen the contribution.
- LaMPlace (ICLR'25) uses pre-route STA labels and macro-only features; PPASurrogate's label study shows this supervision is low-fidelity, and mixed-size features yield higher rank accuracy—ablation results credibly support this claim.
- Prior RL and diffusion-based macro placement methods (AlphaChip, ChiPFormer, diffusion) targeted HPWL/routability proxies; by focusing on post-route PPA, this paper helps bridge the proxy-to-PPA gap highlighted in recent surveys and benchmarks.

Discussion of broader impact and significance

- The demonstrated PPA gains over a hierarchical macro placement baseline are noteworthy for practitioners, suggesting that surrogate-guided configuration tuning is a viable, low-cost lever for timing closure.
- The label fidelity analysis offers a reusable guideline for future cross-stage predictors in placement and beyond (e.g., congestion or power prediction), with practical runtime considerations.
- Potential risks include over-reliance on a surrogate trained at one node/tool flow or design distribution; domain shift mitigation (e.g., uncertainty modeling, periodic ground-truth checks) would make the approach more robust for industrial deployment.
- Openness (code/data) would accelerate community uptake and facilitate standardized comparisons.

Questions for Authors

1. How exactly is the multi-objective prediction vector scalarized within BO? Do you use a weighted sum, lexicographic ordering, or constraint handling, and how sensitive are results to this choice?
2. Why include Area in the prediction/loss when area is constant per circuit? Does removing Area from the outputs or ranking loss change accuracy or BO outcomes?
3. Can you provide a controlled comparison against AutoDMP (BO with HPWL/RUDY proxies) under the same 50-trial budget, on the same five held-out circuits?
4. What are the offline training data generation and training costs in detail (machines, parallelism, total GPU/CPU hours) for the 10-circuit, ~5,000-sample dataset? How would this scale to more designs or advanced nodes?
5. Did you evaluate uncertainty estimation or ensemble predictors to improve robustness during BO (e.g., avoid surrogate misguidance)? If tried, how did this affect convergence and final PPA?
6. How robust are the selected configurations across placement seeds on all five test circuits (not just bp_be)? Please report mean±std post-route WNS/TNS over multiple seeds per circuit.
7. Could a multi-fidelity BO variant (occasional post-GRT validation inside the loop) further improve outcomes or reduce reliance on surrogate accuracy? Any preliminary results?
8. For cross-placer transfer, have you explored lightweight fine-tuning on a handful of RTLMP samples to narrow the accuracy gap, and if so, what is the data/compute required and the achieved improvement?

Overall Assessment

This paper addresses a timely and important problem: bridging the well-documented gap between wirelength-oriented placement objectives and true post-route PPA. Its two main contributions—a rigorous label-fidelity study

establishing post-GRT as the right supervision target, and a mixed-size dual-stream surrogate deployed in a placer-agnostic BO loop—are well-motivated and experimentally supported. The reported timing improvements over a strong hierarchical baseline with modest trial budgets, plus cross-placer transfer, suggest both scientific and practical value.

There are, however, areas to strengthen before publication. The scalarization used for multi-objective BO needs to be specified and justified; a direct comparison to AutoDMP would clarify the advantage over proxy-based BO; and the offline data generation/training costs should be reported more transparently. Addressing these points would solidify the paper's claims and reproducibility. On balance, I find the contributions significant and the empirical evidence compelling enough to merit acceptance at ICCAD, contingent on clarifications and minor additions.

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